
The Science that wins Matches
Learning physics through play
F = ma
Aarav Jain

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FOREWARD

For many young students, physics enters their lives as a collection of formulas on a black board, some hand-wavy popular anecdotes, some keywords in some social videos and more so - some problems to be solved to optimize for marks! Too often it strips the joy of calculation for curiosity which is replaced by caution. The nuances in everyday life and physics application is replaced by pressure.

But physics was never meant to begin that way.

Long before equations, physics begins with questions. These questions suspiciously sound like play: Why does the ball curve? Why does it slow down? Why does jumping feel different on sand than on concrete? These are not “simpler” questions. They are the right ones, which on mulling over explore laws which can explain motion at all classical scales and examples.

This book starts exactly there.

I came to this work as someone who has spent years thinking about physics at its most abstract and equal years watching young students struggle to see why it matters. Aarav Jain does not ask the reader to first believe in physics and then apply it. He invites them to notice, to try, and to feel and only then to name what is happening. What this book does particularly well is resist two common traps. The first is oversimplification: the temptation to dilute ideas until they lose their truth. The second is intimidation: presenting correct physics in a way that feels inaccessible. Here, concepts like Newton’s laws, energy, momentum, projectile motion, drag, and even the Magnus effect are introduced without distortion and importantly, without fear

This book is for students who think science is “not for them,” and for those who love sport but have never been told that their instincts already contain physics. It is also for educators and parents who believe that learning works best when the body and mind are not treated as separate.

If this book helps even a few readers realise that physics is not something to be feared, but something they have been interacting with all along, it will have done something genuinely valuable.

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Why Does The Ball Move The Way It Does?

Why does a football keep rolling after you kick it?
Why is it harder to push something heavy?
Why does the ball bounce back from a wall?

Newton's Laws of Motion explain how things move in the world around us.

The first law- Inertia says that objects like to keep doing what they're already doing, if something is still, it stays still, and if it's moving, it keeps moving, unless something else pushes or pulls it. That's why a ball doesn't start rolling on its own and why a moving car keeps going until the brakes are applied.

The second law- Force and Mass explains how motion changes. It tells us that the harder you push something, the faster it will move, and that heavier objects need more force to move than lighter ones. This is why it's easy to kick a soccer ball far, but much harder to push a car.



The third law- Action and Reaction says that forces always come in pairs. Whenever you push on something, it pushes back on you just as hard. That's why you move forward when you jump off the ground, or why a rocket can launch by pushing exhaust gases downward.

Together, these laws help us understand everything from walking and driving to sports and space travel.

PHYSICS IN 10 SECONDS

- Motion follows simple rules
- Friction slows things down
- Mass affects how fast things speed up
- Pushes always come in pairs



COACH'S TIP

- Stronger kicks = more force
- Smooth passes travel farther
- Walls and rebounds use action-reaction

Football isn't just skill- ***it's physics at play.***

TRY IT YOURSELF!

MATERIALS YOU'LL NEED:

- 1-2 footballs
- 2 plastic bottles - 1 empty,
  1 filled with sand/water
- Chalk / cones
- Open space (about 10m)

SETUP:

1. Mark a 10-metre lane with a start and finish line.
2. Place the empty bottle and the filled bottle at the start.
3. Make sure the path is clear.

ACTIVITY 1
INERTIA RELAY (FIRST LAW)

Question: What happens if nothing stops the ball?

What to do:

1. Gently roll a football on smooth ground.
2. Watch it move.
3. Try stopping it with one finger.
4. Roll the ball again on grass.

What you'll notice:

- The ball keeps rolling on smooth ground
- It slows faster on grass

Why?

- Smooth ground → less friction
- Grass → more friction slowing it down



ACTIVITY 2:
HEAVY VS LIGHT PUSH (SECOND LAW)

Question:
Does weight change how fast something moves?

What to do:

- Push the empty bottle from start to finish.
- Count the time aloud.
- Push the filled bottle with the same effort.
- Count again.

What you'll notice:

- The empty bottle moves faster
- The filled bottle is harder to start and stop

Why?

- Heavier objects need more force for the same acceleration



ACTIVITY 3:
WALL PASS (THIRD LAW)

Question:
Does the ball push back?

What to do:

- Gently kick the ball at a sturdy wall.
- Watch it bounce back.
- Notice the feeling in your foot.
- Push your palms against the wall.

What you'll notice:

- The ball comes back
- The wall pushes back on your hands

Why?

- Every action has an equal and opposite reaction



The Invisible Opponent

Why does every jump come back down — no matter how strong the athlete is?

Gravity- Athletes can fight defenders, wind, and water. But no one can beat gravity.

Gravity explains why objects fall back to the ground and why every jump, throw, or dive has a highest point.

Gravity is a force that pulls everything toward the Earth. It acts on all objects all the time and always pulls straight downward. Even when a ball is moving upward, gravity is already pulling it back toward the ground.

When you jump or throw something, your muscles push you upward, but gravity immediately starts slowing you down. As you rise, gravity reduces your upward speed until, at the highest point, your motion becomes zero. After that, gravity speeds you up downward, bringing you back to the ground.

How high you jump or how far you throw depends on how hard you push
off, your body position, and gravity. While effort and technique matter, gravity always stays the same near Earth and sets a limit on how high things can go.

Together, these ideas help us understand jumping, throwing, falling, and why every athlete eventually comes back down to the ground.

PHYSICS IN 10 SECONDS

- Gravity acts all the time
- It slows things going up and speeds them up going down
- Height is limited by gravity



COACH'S TIP

Jump higher by improving technique, not by fighting gravity.

Gravity always wins — smart athletes work *with* it.

TRY IT YOURSELF!

MATERIALS YOU'LL NEED:

- Chalk to mark jump height
- Measuring tape or wall marks
- A ball (any kind)

ACTIVITY 1
JUMP AND MARK

Question:
How high can you really go?

What to do:

1. Stand next to a wall.
2. Jump straight up and touch the wall.
3. Mark the highest point.
4. Try again with a stronger jump.

What you'll notice:

- Better push = higher mark
- But everyone comes back down



ACTIVITY 2:
THROW UP, WATCH DOWN

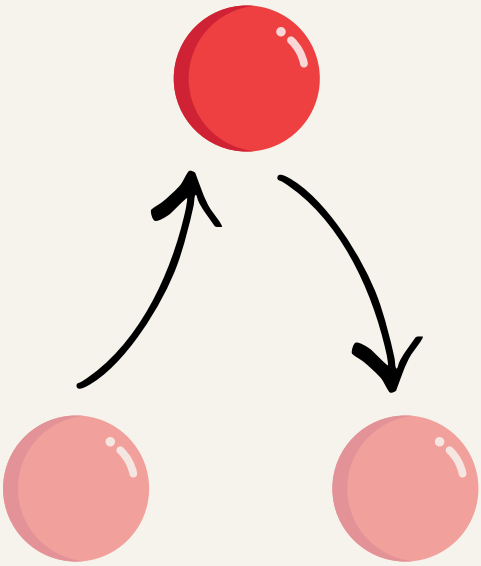
Question:
What happens at the top?

What to do:

- Toss a ball straight upward.
- Watch carefully near the highest point.

What you'll notice:

- The ball slows down
- Stops for a moment
- Then comes back faster



The Fuel Behind Every Move

Why does each jump get lower — even when you keep trying just as hard?

The answer is **energy.** Energy is what allows your body to move, jump, run, and play. Without energy, no movement is possible.

Energy comes in different forms. When you are moving, you have **kinetic energy**. When energy is stored because of height or position, it is called **potential energy**. Your body is constantly changing one kind of energy into another.

When you jump, your muscles first use stored chemical energy. As you push off the ground, this energy becomes kinetic energy as your body moves upward. At the highest point of the jump, your motion slows down and the energy is stored as potential energy.

When you land, the energy changes again. Some of it becomes heat in your muscles, some becomes sound when your feet hit the ground, and some spreads as vibrations through the ground. This energy does not disappear, but it becomes spread out and harder to use again.

Because some energy is lost each time it changes form, every jump becomes a little lower unless new energy is added.

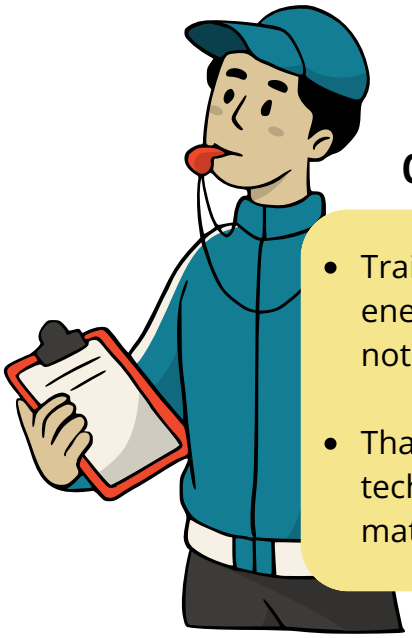
- Moving energy → kinetic energy
- Stored energy → potential energy

Your body constantly changes one into the other.

Together, these ideas explain why athletes get tired, why repeated jumps lose height, and why energy is the fuel behind every move.

PHYSICS IN 10 SECONDS

- Moving = kinetic energy
- Height = potential energy
- Energy spreads out and is lost
- Energy changes form but is partly lost
- Less useful energy means lower jumps



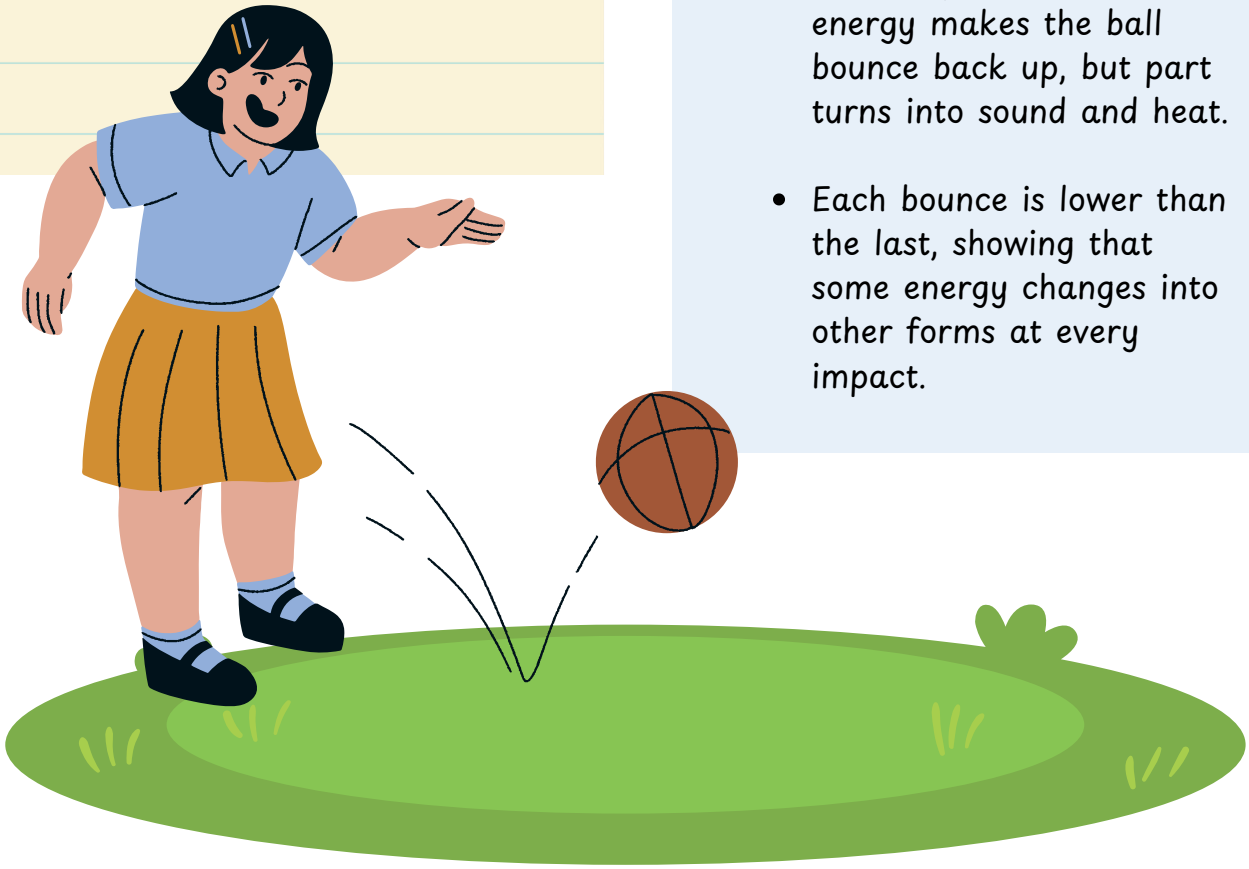
COACH'S TIP

- Train to use energy efficiently, not just forcefully.
- That's why technique and rest matter.

TRY IT YOURSELF!

MATERIALS YOU'LL NEED:

- A ball



ACTIVITY 1
BOUNCING BALL

Question: Why doesn't the ball bounce forever?

What to do:

1. Drop a ball from shoulder height.
2. Watch each bounce.
3. Compare bounce height on hard vs soft ground.

What you'll notice:

- As you bounce the ball it falls faster as potential energy turns into kinetic energy. When it hits the ground, part of that energy makes the ball bounce back up, but part turns into sound and heat.
- Each bounce is lower than the last, showing that some energy changes into other forms at every impact.

ACTIVITY 2:
JUMP REPEATS

Question:
Can you jump the same height every time?

What to do:

- Jump and touch a wall mark.
- Jump again immediately.
- Repeat several times.

What you'll notice:

- Jump height slowly decreases
- Muscles get tired



The Big Question

Why does a fast swing send the ball flying, and why does a softer hit hurt less?

Every time a bat hits a ball, more than strength is involved. Three physics ideas work together during every hit: **momentum, force,** and **impulse**.

Momentum tells us how hard it is to stop something that is moving. A faster bat has more momentum than a slower one, and a heavier ball needs more momentum to be sped up. When the bat has more momentum, it can transfer more motion to the ball, sending it farther. This is why good hitters focus on bat speed, not just muscle power.

Momentum = mass × speed

Impulse explains what happens during the short moment when the bat and ball touch. It depends on how strong the force is and how long the contact lasts. A very quick, hard hit uses a large force over a tiny amount of time. A softer hit spreads the same change in motion over a longer time. Because the force is spread out, the hit feels gentler on your hands.This is why padded gloves, softer bat faces, and controlled swings reduce the “sting” even when the ball still moves forward.

Impulse = force × time

How a shot turns out depends on several factors. Faster bat speed increases momentum. Hitting the ball on the sweet spot transfers motion more efficiently. Choking up on the bat improves control but slightly reduces power. Heavier balls are harder to speed up and usually travel less far with the same swing.

Together, momentum, force, and impulse explain why smart hitters don’t just swing harder — they swing better.

PHYSICS IN 10 SECONDS

- Faster bat = more momentum
- Heavier ball = harder to speed up
- Longer contact time = smaller force and less pain



COACH'S TIP

- Power is not just strength. It's **speed, timing,** and **clean contact.**
- Good hitters don't just swing harder. They swing **faster**, hit the **sweet spot,** and **control** the bat

TRY IT YOURSELF!

MATERIALS YOU'LL NEED:

- 1 bat
- 2 tennis balls (one lightly taped to make it heavier)
- Chalk (to mark distances)
- A folded towel + rubber bands



ACTIVITY 1
BAT SPEED & DISTANCE

Question: Does swing speed matter?

Steps:

1. Use the same tennis ball.
2. Take three gentle swings and hit the ball.
3. Mark where the ball lands with chalk.
4. Take three fast swings.
5. Compare the average distance.

What you'll see:
Fast swings send the ball farther.

Why?
A faster bat has more momentum to pass to the ball.

ACTIVITY 2
BALL MASS

Question: Does a heavier ball fly as far?

Steps:

1. Keep your swing speed the same.
2. Hit the normal tennis ball.
3. Hit the taped (heavier) tennis ball.
4. Mark and compare distances.

What you'll see:
The heavier ball usually goes less far.

Why?
Heavier balls need more momentum to speed up.
With the same swing, they can't move as fast.

ACTIVITY 3:
SOFT VS HARD CONTACT (IMPULSE)

Question:
Why does a softer hit hurt less?

Steps:

1. Wrap the folded towel around the bat face.
2. Secure it with rubber bands.
3. Have someone gently toss the ball.
4. Hit the ball and notice how it feels.

What you'll notice:

- Less sting in your hands
- The ball moves slower

Why?
The towel makes the contact last longer. The force is spread out, so it feels softer.

The Science of the Kick

Why does a ball kicked at the right-angle travel farther than one kicked too low or too high?

When you kick a ball, it doesn't move in just one way. It moves forward and upward at the same time. This kind of movement is called **projectile motion**, and it explains why the angle of a kick matters so much.

A kicked ball has two kinds of motion happening together. One is forward motion, which decides how far the ball travels. The other is up-and-down motion, which decides how long the ball stays in the air. While the ball is moving, gravity is always pulling it back toward the ground.

The angle of the kick decides how your effort is shared between these two motions. A low kick sends the ball forward quickly but keeps it in the air for only a short time. A very high kick sends the ball upward but does not move it forward much. A kick at the right angle gives the ball enough forward speed and enough time in the air to travel the farthest.

On flat ground, this balance usually happens at an angle close to **45 degrees.**

Together, forward motion and upward motion explain why smart kicks travel farther than powerful but poorly angled ones.

PHYSICS IN 10 SECONDS

- A kicked ball moves forward and upward at the same time
- Gravity pulls it down
- Angle and speed decide how far it travels



COACH'S TIP

Great long passes and shots aren't just about power.
They're about choosing the right angle.

- The forward push makes the ball go far
- The upward push keeps the ball in the air
- The best kick uses both, not just one

A kick at about 45° gives the best mix, so the ball goes farthest.

TRY IT YOURSELF!

MATERIALS YOU'LL NEED:

- 1 football
- Chalk or leaves to mark landing spots
- A cardboard angle gate (L-shape marked at ~15°, 30°, 45°)
- Measuring tape or steps

SETUP:

1. Draw a kicking line on the ground.
2. Place the angle gate in front of the ball.
3. Use the gate to help aim the kick.
4. Make sure the landing area is clear.

ACTIVITY 1
ANGLE TRIALS

Question:
Which angle sends the ball the farthest?

What to do:

1. Kick the ball three times at about 15°.
2. Mark each landing spot.
3. Repeat for about 30°.
4. Repeat for about 45°.
5. Measure the distance using steps or a tape.

What you'll notice:

- Kicks near 45° usually travel the farthest.

ACTIVITY 2
SAME KICKER, DIFFERENT ANGLE

Question: Does angle matter more than effort?

What to do:

1. Use the same kicker each time.
2. Keep the effort medium.
3. Change only the angle of the kick.
4. Compare distances.

What you'll notice:

- Changing the angle alone can change the distance a lot.

Try This Too! (Extensions)

- Compare a low drive and a chip shot.
- Which stays in the air longer? Which goes farther?
- Try the same kicks on a small uphill or downhill.
- See how gravity changes the result.



How Air and Spin Bend the Game

Why does a spinning ball curve in the air — and why does it turn sharply after bouncing?

The answer lies in a physics idea called the **Magnus effect.**

When a ball spins, it does not move through the air evenly. Different parts of the ball move through the air at different speeds, and this changes how the air pushes on it.

As the ball spins, one side of the ball moves in the same direction as the air flowing past it, while the other side moves against the air. Because of this, air moves faster on one side of the ball and slower on the other.

Faster-moving air creates lower pressure, while slower-moving air creates higher pressure. The higher pressure pushes the ball sideways, causing it to curve. This sideways push is called the **Magnus force**, and it is what bends the ball's path through the air.

What the Magnus Effect Can Do

- Curve a football free kick
- Make a tennis topspin dip
- Help a cricket ball turn or drift

The faster the spin, the stronger the curve.

Spin vs Swing (Not the Same Thing!)

Spin (Magnus Effect)

- Caused by fingers and wrist
- Works because the ball is spinning
- Makes the ball curve in the air and after the bounce

Swing

- Caused by seam and surface
- Works even without spin
- Makes the ball drift only in the air

Together, air and rotation explain how skilled players make the ball bend, dip, and surprise their opponents.

PHYSICS IN 10 SECONDS

- Spinning balls bend because of air pressure
- Faster spin = stronger Magnus effect
- Control and technique matter more than speed



COACH'S TIP

- Clear finger action creates spin
- Medium pace helps air effects show up
- Smart bowlers use physics to surprise batters

That's the Magnus effect winning wickets.

TRY IT YOURSELF!

MATERIALS YOU'LL NEED:

- 2–3 tennis balls
- Electrical or cloth tape
- Marker (to draw a seam)
- Chalk stumps



SETUP:

1. Mark a bowling line and set up stumps.
2. Prepare two balls:
   - Spin ball: normal tennis ball
   - Swing ball: tape half the ball and draw a seam
3. Keep the landing area clear.

ACTIVITY 1
SPIN BOWLING (MAGNUS EFFECT)

Question: How does spin change the ball's path?

What to do:

1. Show your fingers clearly before bowling.
2. Try off-spin (right-hander: fingers roll right → left).
3. Try leg-spin (fingers roll left → right).
4. Bowl at medium speed and watch: Curve in the air |Turn after the bounce

What you'll notice:

- The ball curves slightly before landing
- It turns strongly after bouncing

ACTIVITY 2:
SWING BOWLING
(AIR FLOW ONLY)

Question:
Can the ball move without spin?

What to do:

1. Use the taped swing ball.
2. Hold the seam slightly angled.
3. Keep one side smooth and one side rough.
4. Bowl at medium pace.
5. Swap seam direction to try:
    - In-swing
    - Out-swing

What you'll notice:

- The ball drifts sideways in the air
- The movement is small but clear



Why the Pool Pushes Back ?

Why does a swimmer feel water pushing back?
Why does a shuttlecock slow down so quickly?

The answer is simple: **air and water are not empty.** They push back.

Drag — The Push That Slows You Down

Drag is a force that resists motion when something moves through air or water. Whenever you run, throw, swim, or jump, the air or water pushes back against you.

Air creates drag on runners, cyclists, and flying balls. Water creates even more drag because it is thicker than air. This is why moving underwater feels much harder than moving through air.

Drag always works **against motion.** It slows things down, shortens jumps and throws, and makes fast-moving objects lose speed. This is why a shuttlecock slows down much faster than a cricket ball and why swimmers feel resistance with every stroke.



DRAG IN SPORTS

What Increases Drag?	What Reduces Drag?
• Bigger surface area • Rough or uneven shape • Higher speed	• Smooth shapes • Streamlined positions • Pointed or curved fronts
That's why: • Shuttlecocks slow down quickly • Parachutes fall slowly • Cyclists wear tight suits	That's why: • Swimmers stretch out straight • Fast balls are smooth • Racing cars look sleek





PHYSICS IN 10 SECONDS

- Drag resists motion
- Water pushes harder than air
- Shape and density matter more than weight

COACH'S TIP

- Swimmers reduce drag by staying streamlined
- Fast balls are smooth
- Good athletes work with fluids, not against them



TRY IT YOURSELF!

MATERIALS YOU'LL NEED:

- 2 paper sheets
- 1 tennis ball
- 1 balloon
- Bowl of water
- Stopwatch or counting aloud

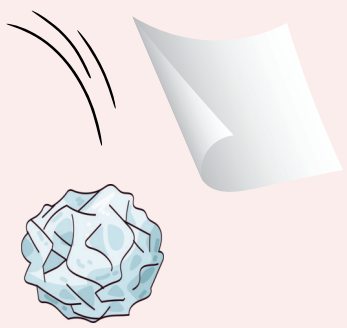
ACTIVITY 1
DRAG IN AIR

Question:
Does shape change how fast something falls?

What to do:
1. Drop a flat paper and a crumpled paper together.
2. Watch which lands first.

What you'll see:
- Crumpled paper falls faster
- Flat paper feels more air pushing back

Why?
More surface → more drag



ACTIVITY 2
DRAG IN WATER

Question:
Why is swimming slow?

What to do:
1. Push a tennis ball slowly through water.
2. Then push it faster.

What you'll feel:
- More pushback at higher speed
- Water drag increases quickly with speed

NOTES

About the Author



Aarav excels in physics, national swimmer and an avid sports person - he is passionate about both sports and science. He believes physics makes the most sense when you can see it, feel it, and try it yourself. Through this book, Aarav hopes to help students discover that physics isn't just a subject in class — it's part of every game they play.

About the Book

For children who dislike reading and who love being outdoors .
For those who love exploring and always question why and how.

The Science That Wins Matches explains core physics ideas using real sports moments that children already love. Through experiments, and easy explanations, readers discover how motion, energy, gravity, and air shape every game.
This book turns abstract physics into something you can feel, test, and play.

Sports based physics explained using simple experiments that can be learned in the Playground. A perfect book for 5th to 8th graders ,written by a student who believes physics should be learned through play, movement, and curiosity, not just textbooks.

